

WILO CLOUD MONITORING

Physics-Guided Data Generation & Extraction Formulation Guide

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1. Physical Reference Model (Wilo Centrifugal Pump-Motor)

The system simulates a industrial Wilo centrifugal pump driven by an 8-pole 3-phase AC induction motor connected to a 50 Hz European mains supply grid. The baseline machine operating parameters are formulated as follows:

1.1 Motor and Shaft Rotation

For an 8-pole induction motor connected to a mains grid frequency of $f_{extline} = 50 \text{ Hz}$, the synchronous shaft speed $N_{extsync}$ (in RPM) is defined by:

$$N_{extsync} = \frac{120 \cdot f_{extline}}{P} = \frac{120 \cdot 50}{8} = 750 \text{ RPM}$$

Accounting for slip under rated operating load ($s \approx 2.67\%$), the nominal shaft speed $N_{extrated}$ is 730 RPM . The fundamental shaft rotational frequency $f_{extshaft}$ (in Hz) is:

$$f_{extshaft} = \frac{N_{extrated}}{60} = \frac{730}{60} \approx 12.17 \text{ Hz}$$

1.2 Defect Frequencies

- Blade-Pass Frequency (BPF):** For a 5-vane centrifugal impeller, the blade-pass rate is:

$$f_{extBPF} = N_{extvanes} \cdot f_{extshaft} = 5 \cdot 12.17 = 60.83 \text{ Hz}$$

- Rolling Element Bearing Defect Frequencies:** Utilizing a 6-ball bearing model with pitch diameter $D = 50 \text{ mm}$, ball diameter $d = 10 \text{ mm}$, and contact angle ($\alpha = 15^\circ$):
 - Ball Pass Frequency Outer Race (BPFO):

$$f_{\text{BPFO}} = \frac{N_b}{2} \cdot f_{\text{shaft}} \cdot \left(1 - \frac{d}{D} \cos \phi\right) = \frac{6}{2} \cdot 12.17 \cdot \left(1 - \frac{10}{50} \cos(15^\circ)\right) \approx 29.45 \text{ Hz}$$

- *Ball Pass Frequency Inner Race (BPFI):*

$$f_{\text{BPFI}} = \frac{N_b}{2} \cdot f_{\text{shaft}} \cdot \left(1 + \frac{d}{D} \cos \phi\right) = \frac{6}{2} \cdot 12.17 \cdot \left(1 + \frac{10}{50} \cos(15^\circ)\right) \approx 43.55 \text{ Hz}$$

- *Ball Spin Frequency (BSF):*

$$f_{\text{BSF}} = \frac{D}{2d} \cdot f_{\text{shaft}} \cdot \left(1 - \left(\frac{d}{D} \cos \phi\right)^2\right) = \frac{50}{20} \cdot 12.17 \cdot \left(1 - \left(0.2 \cos(15^\circ)\right)^2\right) \approx 29.28 \text{ Hz}$$

- *Fundamental Train Frequency (FTF):*

$$f_{\text{FTF}} = \frac{1}{2} \cdot f_{\text{shaft}} \cdot \left(1 - \frac{d}{D} \cos \phi\right) \approx 4.91 \text{ Hz}$$

2. Time-Domain Signal Simulation Models

The three physical channels (Vibration Acceleration $a(t)$, Motor Current $I(t)$, and Acoustic Audio $s(t)$) are simulated as discrete time arrays sampled at $f_s = 700\text{extHz}$ over a duration of $T_d = 2\text{exts}$, yielding $N_s = 1400$ samples.

2.1 Fundamental Physics Models

- **Damped Impact Model (Impulsive Events):** For shocks (e.g., bearing defects, bubble collapse), the response is modeled as a damped sinusoid:

$$x_{\text{impact}}(t) = A \cdot e^{-\zeta \omega_n (t - t_{\text{impact}})} \cdot \sin(\omega_d (t - t_{\text{impact}})) \cdot H(t - t_{\text{impact}})$$

Where $H(t)$ is the Heaviside step function, $\zeta = 0.08$ is the damping ratio, $\omega_n = 2\pi \cdot 1500\text{extrad/s}$ is the resonant frequency, and $\omega_d = \omega_n \sqrt{1 - \zeta^2}$.

- **Randomized Periodic Impact Train:** Impact occurrences are modeled with a repetition period $T = 1/f_{\text{extfault}}$ and timing jitter δ_i :

$$x_{\text{train}}(t) = \sum_i x_{\text{impact}}^{(i)}(t - t_{\text{impact}} = i \cdot T + \delta_i)$$

Where $\delta_i \sim \mathcal{U}(-0.05T, 0.05T)$, and impact amplitude $A_i \sim \text{extLognormal}(\mu, \sigma)$.

2.2 Fault-Specific Signatures

Fault Name	Physical Mechanism & Simulation Formulation
Motor Stall	Rotational speed collapses to 0. Current locks at stator resistance limit: $I(t) = \sqrt{2} \cdot I_{\text{locked}} \cdot \sin(2\pi \cdot 50 \cdot t) + \text{Noise}$, $I_{\text{locked}} \approx 5.5 \cdot I_{\text{rated}}$ Vibration $a(t)$ drops to zero as rotational mechanical force vanishes.
Pump Cavitation	Bubble collapse follows the Rayleigh-Plesset equation, producing explosive bursts: $s_{\text{extcav}}(t) = \sum_k A_k \cdot \text{extdamped_sinusoid}(t, t_k) \quad \text{extclusteredaroundBPFFrate}$ Vibration kurtosis surges. Current exhibits high-frequency load-fluctuation noise.
Impeller Damage	Hydraulic unbalance generates periodic modulations at f_{extshaft} and its harmonics: $a_{\text{extimpeller}}(t) = A_d \cdot (1 + \sin(2\pi f_{\text{extshaft}}t)) \cdot \sin(2\pi f_{\text{extBPF}}t) + \text{extImpacts}$
Shaft Misalignment	Combines angular and parallel unbalance, injecting sharp peaks at f_{extshaft} and $2 \cdot f_{\text{extshaft}}$: $a_{\text{extmis}}(t) = A_1 \sin(2\pi f_{\text{extshaft}}t + \phi_1) + A_2 \sin(2\pi \cdot (2f_{\text{extshaft}})t + \phi_2)$
Motor Winding Failure	Stator short-circuits induce magnetic path distortions, introducing 3rd and 5th current harmonics: $I(t) = I_1 \sin(2\pi \cdot 50 \cdot t) + I_3 \sin(2\pi \cdot 150 \cdot t) + I_5 \sin(2\pi \cdot 250 \cdot t)$ Partial discharge events are modeled as fast, high-voltage impulses: $p(t) \sim \text{extWeibull}(\lambda, k)$.
Motor Overheating	Resistance rises exponentially with temperature $T(t) = T_0 + \Delta T(1 - e^{-t/au})$, resulting in winding current drop. Current harmonics grow slightly. Audio loudness increases linearly.

3. Statistical Feature Formulation

For a discrete-time signal array $X = \{x_1, x_2, \dots, x_N\}$ with mean μ and standard deviation σ , the statistical parameters extracted in the backend are defined as:

- **Mean (μ):** Centroid of the data points.

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

- **Median:** Central value of the sorted sequence.

$$extMedian = extpercentile(X, 50)$$

- **Standard Deviation (σ):** Spread of signal energy.

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

- **Variance (σ^2):** Second statistical moment.

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

- **Range (Peak-to-Peak):** Total span of the waveform.

$$extRange = \max(X) - \min(X)$$

- **Peak (Maximum Absolute Amplitude):** Worst-case structural stress amplitude.

$$extPeak = \max(|X|) = \max(|x_1|, |x_2|, \dots, |x_N|)$$

- **Root Mean Square (RMS):** Total effective power of the signal.

$$extRMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$$

- **Crest Factor (CF):** Shocks indicator (ratio of peak to average energy).

$$CF = \frac{extPeak}{extRMS} = \frac{\max(|X|)}{\sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}}$$

- **Skewness (S):** Measure of symmetry (third standardized moment).

$$S = \frac{1}{N} \sum_{i=1}^N \frac{(x_i - \mu)^3}{\sigma^3}$$

- **Excess Kurtosis (K_e):** Indicator of transient impulsive shocks (Fisher definition where normal distribution $K_e = 0.0$):

$$K_e = \frac{1}{N} \sum_{i=1}^N \frac{(x_i - \mu)^4}{\sigma^4} - 3$$

4. Advanced Diagnostic & Onset Detection Logic

4.1 Fast Fourier Transform (FFT) Expected Frequency Match

The FFT transforms the time-domain signal X to the frequency domain F . The top-5 frequency peaks $f_{extpeaks} = \{f_1, f_2, f_3, f_4, f_5\}$ are evaluated against a set of fault-specific expected frequencies $f_{extexpected}$ with a relative tolerance threshold of $\eta = 10\%$.

The **Fault Frequency Match Score (η)** is the percentage of peak frequencies that fall within expected bands:

$$\eta = \frac{|\{f_{extpeaks} \mid \sum_{f \in f_{extpeaks}} \mathbb{I}(\min_{e \in f_{extexpected}} |f - e| \leq \eta e)\}|}{|f_{extpeaks}|}$$

Where $\mathbb{I}(\cdot)$ is the indicator function returning 1 if true, 0 if false.

4.2 Multi-Sensor CUSUM Onset Detection

For gradual faults, the exact moment of failure onset is identified using a weighted Cumulative Sum (CUSUM) formulation.

- Baseline Calibration:** The mean μ_{bj} and standard deviation σ_{bj} of feature j are calibrated during a healthy baseline window (first 25% of data).
- Z-Score Normalized Deviation:** For interval i , the normalized deviation z_{ij} is calculated:

$$z_{ij} = \frac{x_{ij} - \mu_{bj}}{\sigma_{bj}}$$

- Weighted Deviation Score (D_i):** Fault-specific weights w_j filter direction of change (e.g., current drop on seal leaks, kurtosis rise on bearing wear):

$$D_i = \frac{\sum_j w_j \cdot \max(0, \text{sgn}_j(z_{ij}) \cdot z_{ij} - z_{\text{extmin}})}{\sum_j w_j}$$

Where $z_{\text{extmin}} = 0.8\sigma$ is the sensitivity offset filter, and $\text{extsgn}_j \in \{1, -1, \text{extabsolute}\}$ matches physical direction.

- CUSUM Accumulation (S_i):** The cumulative deviation score accumulates:

$$S_i = \max(0, S_{i-1} + D_i - k)$$

Where $k = 0.5$ is the slack/drift parameter. When $S_i > h$ (decision threshold $h = 3.0$) for 2 consecutive cycles, the deviation onset index is triggered.

4.3 Feature Slope Formulations (Rate of Change)

For each interval index i with timestamp t_i (in milliseconds), the feature rate of change per second is calculated:

- **Forward Difference (for $i < extFailureIndex$):**

$$\text{extSlope}_{i,j} = \frac{x_{i+1,j} - x_{i,j}}{(t_{i+1} - t_i) / 1000}$$

- **Backward Difference (for the terminal $i = extFailureIndex$):**

$$\text{extSlope}_{i,j} = \frac{x_{i,j} - x_{i-1,j}}{(t_i - t_{i-1}) / 1000}$$